Elementary Linear Algebra

Chapter 3

Euclidean Vector Spaces

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Euclidean Vector Spaces

- 3.1 Vectors in 2-Space, 3-Space, and *n*-Space
- 3.2 Norm, Dot Product, and Distance in R^n
- 3.3 Orthogonality
- 3.4 The Geometry of Linear Systems
- 3.5 Cross Product

Section 3.1
Vectors in 2-Space, 3-Space, and *n*-Space

- Geometric Vectors
  - Direction
  - Length

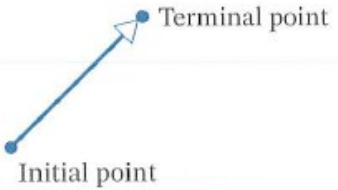


FIGURE 3.1.1

- Equivalent vectors

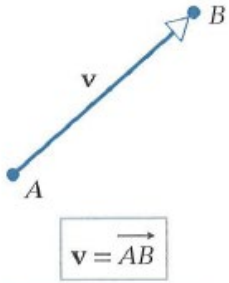


FIGURE 3.1.2

- Zero vector

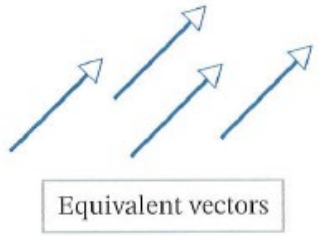


FIGURE 3.1.3

Vector Addition

Parallelogram Rule for Vector Addition

If **v** and **w** are vectors in 2-space or 3-space that are positioned so their initial points coincide, then the two vectors form adjacent sides of a parallelogram, and the ***sum*** **v** + **w** is the vector represented by the arrow from the common initial point of **v** and **w** to the opposite vertex of the parallelogram (**Figure 3.1.4*a***).

Triangle Rule for Vector Addition

If **v** and **w** are vectors in 2-space or 3-space that are positioned so the initial point of **w** is at the terminal point of **v**, then the ***sum*** **v** + **w** is represented by the arrow from the initial point of **v** to the terminal point of **w** (**Figure 3.1.4*b***).

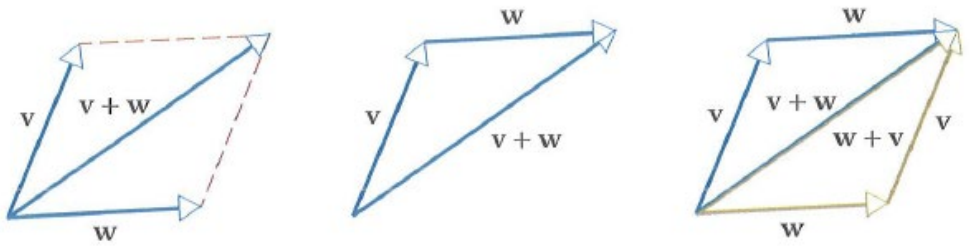


FIGURE 3.1.4

Vector Subtraction

Vector Subtraction

The ***negative*** of a vector $\mathbf{v}$, denoted by $-\mathbf{v}$, is the vector that has the same length as $\mathbf{v}$ but is oppositely directed (**Figure 3.1.6*a***), and the ***difference*** of $\mathbf{v}$ from $\mathbf{w}$, denoted by $\mathbf{w} - \mathbf{v}$, is defined to be the sum

$$\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v}) \tag{2}$$

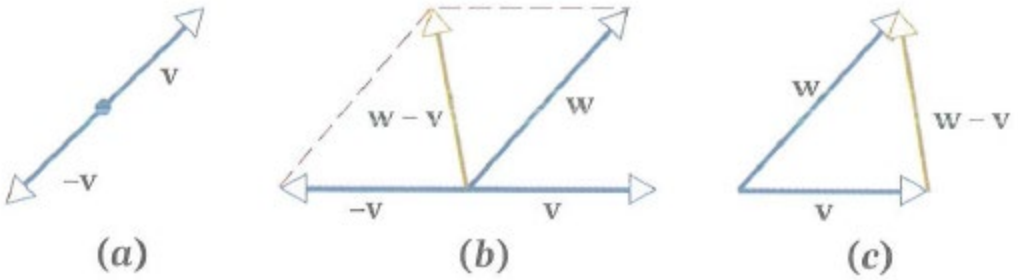


FIGURE 3.1.6

Scalar Multiplication

Scalar Multiplication

If $\mathbf{v}$ is a nonzero vector in 2-space or 3-space, and *if* k is a nonzero scalar, then we define the ***scalar product of*** $\mathbf{v}$ ***by*** $\mathbf{k}$ to be the vector whose length is $|k|$ times the length of $\mathbf{v}$ and whose direction is the same as that of $\mathbf{v}$ if k is positive and opposite to that of $\mathbf{v}$ if k is negative. If $k = 0$ or $\mathbf{v} = \mathbf{0}$, then we define $k\mathbf{v}$ to be $\mathbf{0}$.

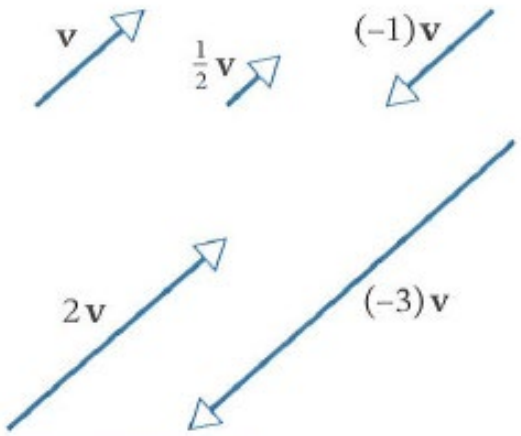


FIGURE 3.1.7

Parallel and Collinear Vectors

- When applied vectors , the terms ‘parallel’ and ‘collinear’ mean the same thing

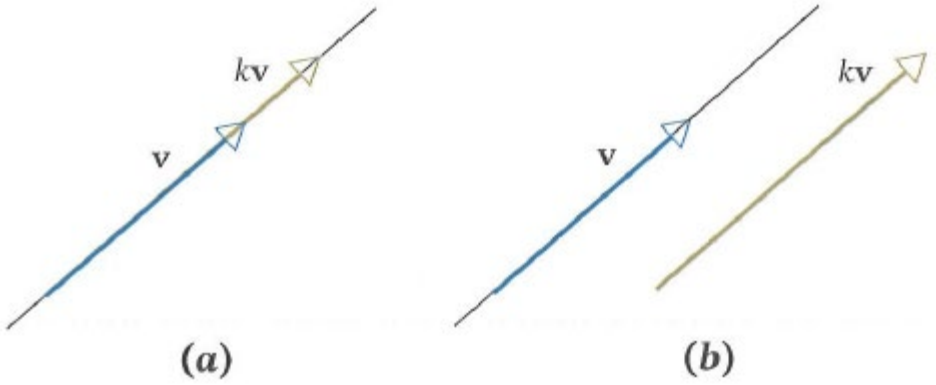


FIGURE 3.1.8

Sums of Three or More Vectors

- Associative law for addition
  - It does not matter which two we add first

$$\mathbf{u}+(\mathbf{v}+\mathbf{w})=(\mathbf{u}+\mathbf{v})+\mathbf{w}$$

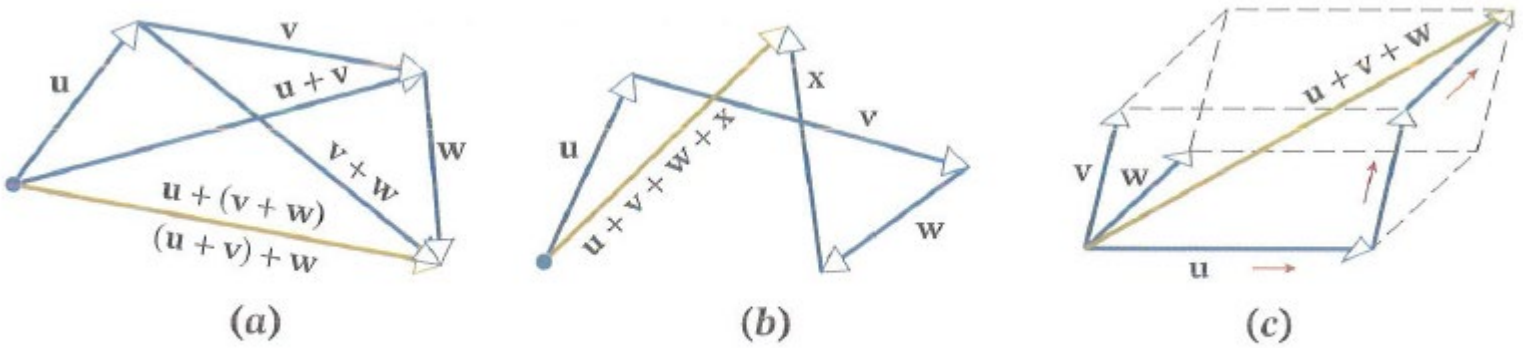


FIGURE 3.1.9

Vectors in Coordinate Systems

- Rectangular coordinate system with the vector's initial point at the origin
  - Vector in 2-space
  - Vector in 3-space

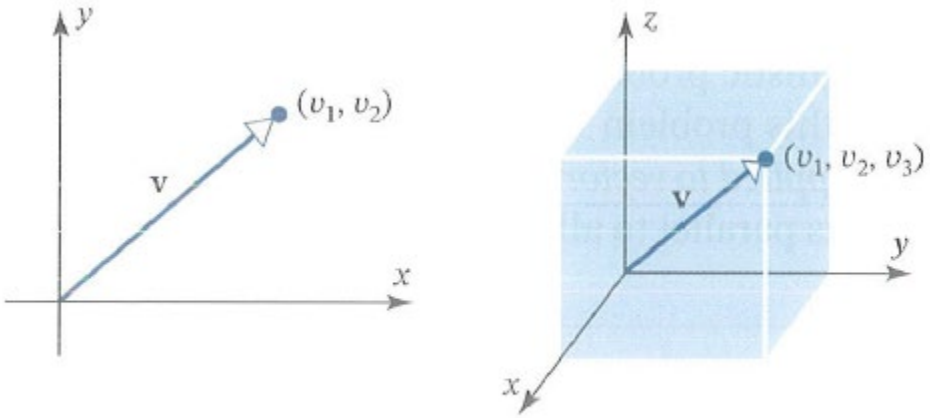


FIGURE 3.1.10

Vectors Whose Initial Point Is Not at the Origin

- When the initial points of the two vectors are not at the origin, then the components of this vector are given by
  - In 2-Space

$$\overrightarrow{P_1P_2} = (x_2 - x_1, y_2 - y_1)$$

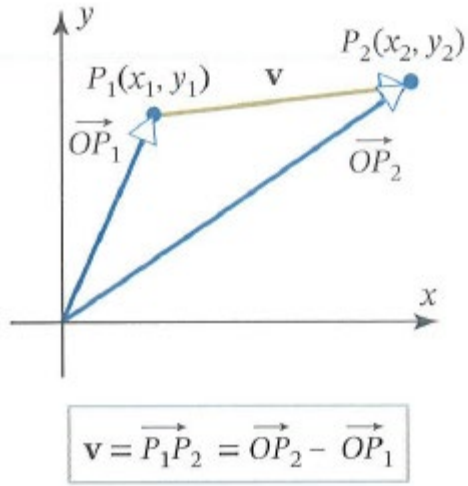


FIGURE 3.1.12

  - In 3-Space

$$\overrightarrow{P_1P_2} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

EXAMPLE 1 | Finding the Components of a Vector

The components of the vector $\mathbf{v} = \overrightarrow{P_1P_2}$ with initial point $P_1(2, -1, 4)$ and terminal point $P_2(7, 5, -8)$ are

$$\mathbf{v} = (7 - 2, 5 - (-1), (-8) - 4) = (5, 6, -12)$$

n-Space

Definition 1

If n is a positive integer, then an ***ordered n-tuple*** is a sequence of n real numbers $(v_1, v_2, \ldots, v_n)$. The set of all ordered n-tuples is called ***real n-space*** and is denoted by R^n.

- Typical applications that lead to *n*-tuples
  - Experimental data
  - Storage and Warehousing
  - Electrical circuits
  - Graphical images
  - Economics
  - Mechanical systems

Operations on Vectors in R^n

Definition 2

Vectors $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ and $\mathbf{w} = (w_1, w_2, \ldots, w_n)$ in R^n are said to be ***equal*** (also called ***equivalent***) if

$$v_1 = w_1, \quad v_2 = w_2, \ldots, \quad v_n = w_n$$

We indicate this by writing $\mathbf{v} = \mathbf{w}$.

EXAMPLE 2 | Equality of Vectors

The vectors

$$\mathbf{v} = (a, b, c, d) \quad \text{and} \quad \mathbf{w} = (1, -4, 2, 7)$$

are equal if and only if $a = 1, b = -4, c = 2$, and $d = 7$.

Definition 3

If $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ and $\mathbf{w} = (w_1, w_2, \ldots, w_n)$ are vectors in R^n, and if k is any scalar, then we define

$$\mathbf{v} + \mathbf{w} = (v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n) \tag{10}$$

$$k\mathbf{v} = (kv_1, kv_2, \ldots, kv_n) \tag{11}$$

$$-\mathbf{v} = (-v_1, -v_2, \ldots, -v_n) \tag{12}$$

$$\mathbf{w} - \mathbf{v} = \mathbf{w} + (-\mathbf{v}) = (w_1 - v_1, w_2 - v_2, \ldots, w_n - v_n) \tag{13}$$

EXAMPLE 3 | Algebraic Operations Using Components

If $\mathbf{v} = (1, -3, 2)$ and $\mathbf{w} = (4, 2, 1)$, then

$$\mathbf{v} + \mathbf{w} = (5, -1, 3), \qquad 2\mathbf{v} = (2, -6, 4)$$

$$-\mathbf{w} = (-4, -2, -1), \qquad \mathbf{v} - \mathbf{w} = \mathbf{v} + (-\mathbf{w}) = (-3, -5, 1)$$

- Summary of the most important properties of vectors operations

Theorem 3.1.1

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in R^n, and if k and m are scalars, then:

(a) $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$

(b) $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$

(c) $\mathbf{u}+\mathbf{0}=\mathbf{0}+\mathbf{u}=\mathbf{u}$

(d) $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$

(e) $k(\mathbf{u}+\mathbf{v})=k\mathbf{u}+k\mathbf{v}$

(f) $(k+m)\mathbf{u}=k\mathbf{u}+m\mathbf{u}$

(g) $k(m\mathbf{u})=(km)\mathbf{u}$

(h) $1\mathbf{u}=\mathbf{u}$

Theorem 3.1.2

If $\mathbf{v}$ is a vector in R^n and k is a scalar, then:

(a) $0\mathbf{v}=\mathbf{0}$

(b) $k\mathbf{0}=\mathbf{0}$

(c) $(-1)\mathbf{v}=-\mathbf{v}$

Linear Combinations

Definition 4

If $\mathbf{w}$ is a vector in R^n, then $\mathbf{w}$ is said to be a ***linear combination*** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r$ in R^n if it can be expressed in the form

$$\mathbf{w} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \cdots + k_r\mathbf{v}_r \tag{14}$$

where $k_1, k_2, \ldots, k_r$ are scalars. These scalars are called the ***coefficients*** of the linear combination. In the case where $r = 1$, Formula (14) becomes $\mathbf{w} = k_1\mathbf{v}_1$, so that a linear combination of a single vector is just a scalar multiple of that vector.

Section 3.2
Norm, Dot Product, and Distance in R^n

- Norm of a vector
  - The length of a vector in 2-space

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}$$

  - The length of a vector in 3-space

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

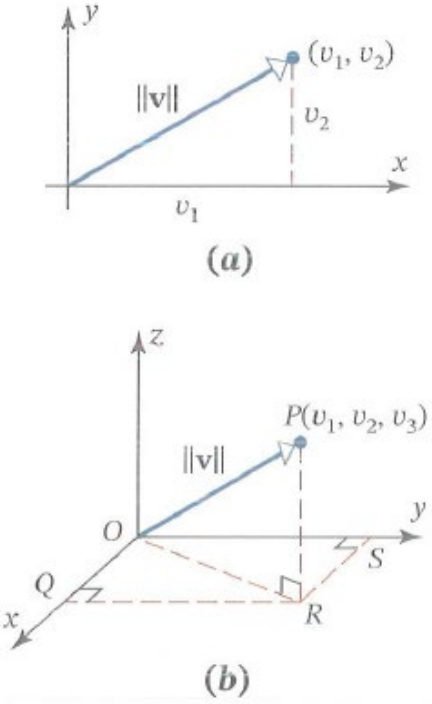


FIGURE 3.2.1

Definition 1

If $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ is a vector in R^n, then the ***norm*** of $\mathbf{v}$ (also called the ***length*** of $\mathbf{v}$ or the ***magnitude*** of $\mathbf{v}$) is denoted by $\|\mathbf{v}\|$, and is defined by the formula

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \qquad (3)$$

EXAMPLE 1 | Calculating Norms

It follows from Formula (2) that the norm of the vector $\mathbf{v} = (-3, 2, 1)$ in R^3 is

$$\|\mathbf{v}\| = \sqrt{(-3)^2 + 2^2 + 1^2} = \sqrt{14}$$

and it follows from Formula (3) that the norm of the vector $\mathbf{v} = (2, -1, 3, -5)$ in R^4 is

$$\|\mathbf{v}\| = \sqrt{2^2 + (-1)^2 + 3^2 + (-5)^2} = \sqrt{39}$$

Theorem 3.2.1

If $\mathbf{v}$ is a vector in R^n, and if k is any scalar, then:

(a) $\|\mathbf{v}\| \geq 0$

(b) $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$

(c) $\|k\mathbf{v}\| = |k|\|\mathbf{v}\|$

Unit Vectors

- Unit vector : A vector of norm 1

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|}\mathbf{v}$$

- Normalizing : The process of multiplying a nonzero vector by the reciprocal of its length to obtain a unit vector

EXAMPLE 2 | Normalizing a Vector

Find the unit vector $\mathbf{u}$ that has the same direction as $\mathbf{v} = (2, 2, -1)$.

Solution The vector $\mathbf{v}$ has length

$$\|\mathbf{v}\| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$$

Thus, from (4)

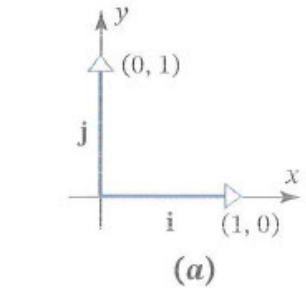
$$\mathbf{u} = \tfrac{1}{3}(2, 2, -1) = \left(\tfrac{2}{3}, \tfrac{2}{3}, -\tfrac{1}{3}\right)$$

As a check, you may want to confirm that $\|\mathbf{u}\| = 1$.

The Standard Unit Vectors

- Standard unit vectors : The unit vectors in the positive directions of the coordinate axes
  - In 2-space,

$$\mathbf{i} = (1,0) \quad \text{and} \quad \mathbf{j} = (0,1)$$



  - In 3-space,

$$\mathbf{i} = (1,0,0),\ \mathbf{j} = (0,1,0), \quad \text{and} \quad \mathbf{k} = (0,0,1)$$

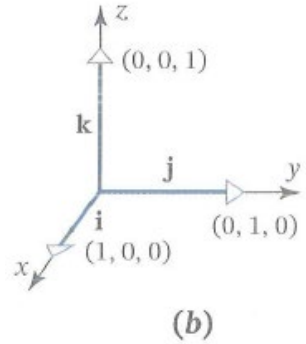


FIGURE 3.2.2

  - In R^n,

$$\mathbf{e}_1 = (1,0,0,\cdots,0),\ \mathbf{e}_2 = (0,1,0,\cdots,0),\ \cdots,\ \mathbf{e}_n = (0,0,0,\cdots,1)$$

EXAMPLE 3 | Linear Combinations of Standard Unit Vectors

$$(2, -3, 4) = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$$
$$(7, 3, -4, 5) = 7\mathbf{e}_1 + 3\mathbf{e}_2 - 4\mathbf{e}_3 + 5\mathbf{e}_4$$

Distance in R^n

- Distance in 2-Space

$$d = \left\| \overrightarrow{P_1P_2} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

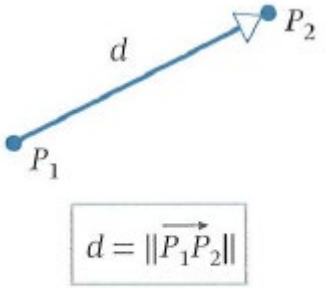


FIGURE 3.2.3

- Distance in 3-Space

$$d(\mathbf{u}, \mathbf{v}) = \left\| \overrightarrow{P_1P_2} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Definition 2

If $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ are points in R^n, then we denote the ***distance*** between $\mathbf{u}$ and $\mathbf{v}$ by $d(\mathbf{u}, \mathbf{v})$ and define it to be

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2} \qquad (11)$$

EXAMPLE 4 | Calculating Distance in R^n

If

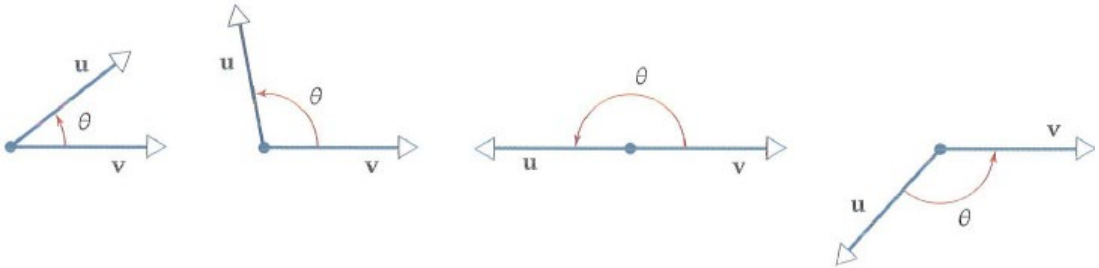
$$\mathbf{u} = (1, 3, -2, 7) \quad \text{and} \quad \mathbf{v} = (0, 7, 2, 2)$$

then the distance between $\mathbf{u}$ and $\mathbf{v}$ is

$$d(\mathbf{u}, \mathbf{v}) = \sqrt{(1-0)^2 + (3-7)^2 + (-2-2)^2 + (7-2)^2} = \sqrt{58}$$

Dot Product

- Angle between **u** and **v**



The angle θ between **u** and **v** satisfies $0 \le \theta \le \pi$.

FIGURE 3.2.4

- Dot product

Definition 3

If **u** and **v** are nonzero vectors in R^2 or R^3, and if θ is the angle between **u** and **v**, then the ***dot product*** (also called the ***Euclidean inner product***) of **u** and **v** is denoted by $\mathbf{u} \cdot \mathbf{v}$ and is defined as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta \qquad (12)$$

If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then we define $\mathbf{u} \cdot \mathbf{v}$ to be 0.

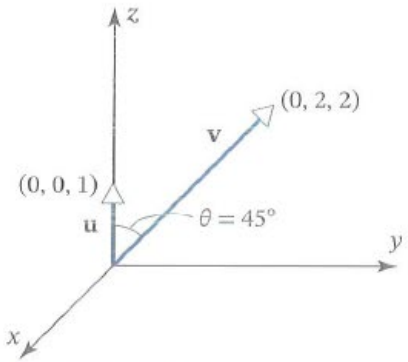


FIGURE 3.2.5

EXAMPLE 5 | Dot Product

Find the dot product of the vectors shown in **Figure 3.2.5**.

Solution The lengths of the vectors are

$$\|\mathbf{u}\| = 1 \quad \text{and} \quad \|\mathbf{v}\| = \sqrt{8} = 2\sqrt{2}$$

and the cosine of the angle θ between them is

$$\cos(45°) = 1/\sqrt{2}$$

Thus, it follows from Formula (12) that

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta = (1)(2\sqrt{2})(1/\sqrt{2}) = 2$$

Component Form of the Dot Product

- Let $\mathbf{u}=(u_1,u_2,u_3)$, $\mathbf{v}=(v_1,v_2,v_3)$, the component form of the dot product is

$$\mathbf{u}\cdot\mathbf{v}=u_1v_1+u_2v_2+u_3v_3$$

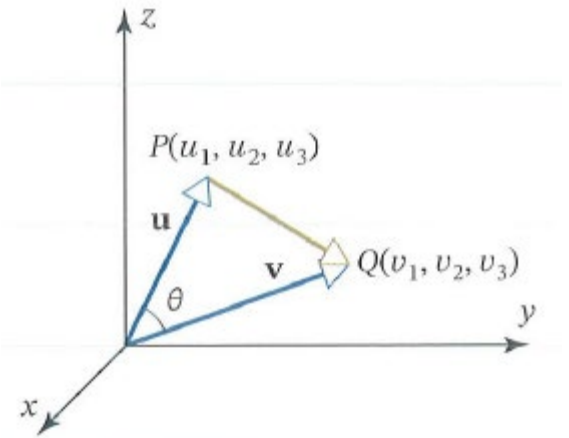


FIGURE 3.2.6

Definition 4

If $\mathbf{u}=(u_1,u_2,\ldots,u_n)$ and $\mathbf{v}=(v_1,v_2,\ldots,v_n)$ are vectors in R^n, then the ***dot product*** (also called the ***Euclidean inner product***) of $\mathbf{u}$ and $\mathbf{v}$ is denoted by $\mathbf{u}\cdot\mathbf{v}$ and is defined by

$$\mathbf{u}\cdot\mathbf{v}=u_1v_1+u_2v_2+\cdots+u_nv_n \qquad (17)$$

EXAMPLE 6 | Calculating Dot Products Using Components

(*a*) Use Formula (15) to compute the dot product of the vectors $\mathbf{u}$ and $\mathbf{v}$ in Example 5.

(*b*) Calculate $\mathbf{u} \cdot \mathbf{v}$ for the following vectors in R^4:

$$\mathbf{u} = (-1, 3, 5, 7), \quad \mathbf{v} = (-3, -4, 1, 0)$$

Solution (*a*) The component forms of the vectors are $\mathbf{u} = (0, 0, 1)$ and $\mathbf{v} = (0, 2, 2)$. Thus,

$$\mathbf{u} \cdot \mathbf{v} = (0)(0) + (0)(2) + (1)(2) = 2$$

which agrees with the result obtained geometrically in Example 5.

Solution (*b*)

$$\mathbf{u} \cdot \mathbf{v} = (-1)(-3) + (3)(-4) + (5)(1) + (7)(0) = -4$$

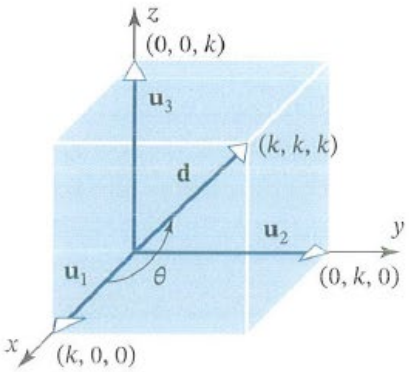


FIGURE 3.2.7

Note that the angle θ obtained in Example 7 does not involve k. Why was this to be expected?

EXAMPLE 7 | A Geometry Problem Solved Using Dot Product

Find the angle between a diagonal of a cube and one of its edges.

Solution Let k be the length of an edge and introduce a coordinate system as shown in Figure 3.2.7. If we let $\mathbf{u}_1 = (k, 0, 0)$, $\mathbf{u}_2 = (0, k, 0)$, and $\mathbf{u}_3 = (0, 0, k)$, then the vector

$$\mathbf{d} = (k, k, k) = \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3$$

is a diagonal of the cube. It follows from Formula (13) that the angle θ between $\mathbf{d}$ and the edge $\mathbf{u}_1$ satisfies

$$\cos\theta = \frac{\mathbf{u}_1 \cdot \mathbf{d}}{\|\mathbf{u}_1\|\|\mathbf{d}\|} = \frac{k^2}{(k)(\sqrt{3k^2})} = \frac{1}{\sqrt{3}}$$

With the help of a calculator we obtain

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \approx 54.74^\circ$$

Algebraic Properties of the Dot Product

Theorem 3.2.2

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in R^n, and if k is a scalar, then:

(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ **[Symmetry property]**

(b) $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ **[Distributive property]**

(c) $k(\mathbf{u} \cdot \mathbf{v}) = (k\mathbf{u}) \cdot \mathbf{v}$ **[Homogeneity property]**

(d) $\mathbf{v} \cdot \mathbf{v} \geq 0$ and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$ **[Positivity property]**

Theorem 3.2.3

If $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in R^n, and if k is a scalar, then:

(a) $\mathbf{0} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{0} = 0$

(b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

(c) $\mathbf{u} \cdot (\mathbf{v} - \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} - \mathbf{u} \cdot \mathbf{w}$

(d) $(\mathbf{u} - \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} - \mathbf{v} \cdot \mathbf{w}$

(e) $k(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (k\mathbf{v})$

EXAMPLE 8 | Calculating with Dot Products

$$
\begin{aligned}
(\mathbf{u} - 2\mathbf{v}) \cdot (3\mathbf{u} + 4\mathbf{v}) &= \mathbf{u} \cdot (3\mathbf{u} + 4\mathbf{v}) - 2\mathbf{v} \cdot (3\mathbf{u} + 4\mathbf{v}) \\
&= 3(\mathbf{u} \cdot \mathbf{u}) + 4(\mathbf{u} \cdot \mathbf{v}) - 6(\mathbf{v} \cdot \mathbf{u}) - 8(\mathbf{v} \cdot \mathbf{v}) \\
&= 3\|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) - 8\|\mathbf{v}\|^2
\end{aligned}
$$

Cauchy-Schwarz Inequality and Angles in R^n

Theorem 3.2.4

Cauchy–Schwarz Inequality

If $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ are vectors in R^n, then

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \, \|\mathbf{v}\| \tag{22}$$

or in terms of components

$$|u_1v_1 + u_2v_2 + \cdots + u_nv_n| \leq (u_1^2 + u_2^2 + \cdots + u_n^2)^{1/2}(v_1^2 + v_2^2 + \cdots + v_n^2)^{1/2} \tag{23}$$

- Simple proof

$$\frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{u}\| \|\mathbf{v}\|} \leq 1$$

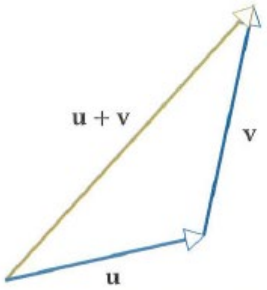
Geometry in R^n

Theorem 3.2.5

If $\mathbf{u}, \mathbf{v}$, and $\mathbf{w}$ are vectors in R^n, then:

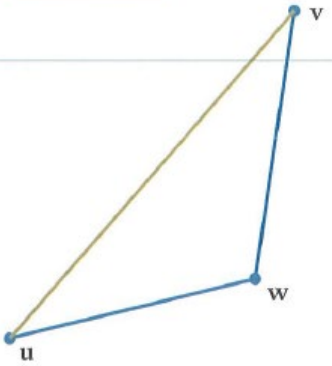
(a) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ **[Triangle inequality for vectors]**

(b) $d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$ **[Triangle inequality for distances]**



$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

FIGURE 3.2.8



$d(\mathbf{u}, \mathbf{v}) \leq d(\mathbf{u}, \mathbf{w}) + d(\mathbf{w}, \mathbf{v})$

FIGURE 3.2.9

Theorem 3.2.6

Parallelogram Equation for Vectors

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n, then

$$\|\mathbf{u}+\mathbf{v}\|^2+\|\mathbf{u}-\mathbf{v}\|^2=2\left(\|\mathbf{u}\|^2+\|\mathbf{v}\|^2\right) \tag{24}$$

Theorem 3.2.7

If $\mathbf{u}$ and $\mathbf{v}$ are vectors in R^n with the Euclidean inner product, then

$$\mathbf{u}\cdot\mathbf{v}=\tfrac{1}{4}\|\mathbf{u}+\mathbf{v}\|^2-\tfrac{1}{4}\|\mathbf{u}-\mathbf{v}\|^2 \tag{25}$$

Dot Products as Matrix Multiplication

- Express the dot product of vectors using matrix notation

TABLE 1

Form	Dot Product	Example	
$\mathbf{u}$ a column matrix and $\mathbf{v}$ a column matrix	$\mathbf{u}\cdot\mathbf{v}=\mathbf{u}^T\mathbf{v}=\mathbf{v}^T\mathbf{u}$	$\mathbf{u}=\begin{bmatrix}1\\-3\\5\end{bmatrix}$ $\mathbf{v}=\begin{bmatrix}5\\4\\0\end{bmatrix}$	$\mathbf{u}^T\mathbf{v}=\begin{bmatrix}1 & -3 & 5\end{bmatrix}\begin{bmatrix}5\\4\\0\end{bmatrix}=-7$ $\mathbf{v}^T\mathbf{u}=\begin{bmatrix}5 & 4 & 0\end{bmatrix}\begin{bmatrix}1\\-3\\5\end{bmatrix}=-7$
$\mathbf{u}$ a row matrix and $\mathbf{v}$ a column matrix	$\mathbf{u}\cdot\mathbf{v}=\mathbf{u}\mathbf{v}=\mathbf{v}^T\mathbf{u}^T$	$\mathbf{u}=\begin{bmatrix}1 & -3 & 5\end{bmatrix}$ $\mathbf{v}=\begin{bmatrix}5\\4\\0\end{bmatrix}$	$\mathbf{u}\mathbf{v}=\begin{bmatrix}1 & -3 & 5\end{bmatrix}\begin{bmatrix}5\\4\\0\end{bmatrix}=-7$ $\mathbf{v}^T\mathbf{u}^T=\begin{bmatrix}5 & 4 & 0\end{bmatrix}\begin{bmatrix}1\\-3\\5\end{bmatrix}=-7$
$\mathbf{u}$ a column matrix and $\mathbf{v}$ a row matrix	$\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\mathbf{u}=\mathbf{u}^T\mathbf{v}^T$	$\mathbf{u}=\begin{bmatrix}1\\-3\\5\end{bmatrix}$ $\mathbf{v}=\begin{bmatrix}5 & 4 & 0\end{bmatrix}$	$\mathbf{v}\mathbf{u}=\begin{bmatrix}5 & 4 & 0\end{bmatrix}\begin{bmatrix}1\\-3\\5\end{bmatrix}=-7$ $\mathbf{u}^T\mathbf{v}^T=\begin{bmatrix}1 & -3 & 5\end{bmatrix}\begin{bmatrix}5\\4\\0\end{bmatrix}=-7$
$\mathbf{u}$ a row matrix and $\mathbf{v}$ a row matrix	$\mathbf{u}\cdot\mathbf{v}=\mathbf{u}\mathbf{v}^T=\mathbf{v}\mathbf{u}^T$	$\mathbf{u}=\begin{bmatrix}1 & -3 & 5\end{bmatrix}$ $\mathbf{v}=\begin{bmatrix}5 & 4 & 0\end{bmatrix}$	$\mathbf{u}\mathbf{v}^T=\begin{bmatrix}1 & -3 & 5\end{bmatrix}\begin{bmatrix}5\\4\\0\end{bmatrix}=-7$ $\mathbf{v}\mathbf{u}^T=\begin{bmatrix}5 & 4 & 0\end{bmatrix}\begin{bmatrix}1\\-3\\5\end{bmatrix}=-7$

$$A\mathbf{u}\cdot\mathbf{v}=\mathbf{u}\cdot A^T\mathbf{v}$$

$$\mathbf{u}\cdot A\mathbf{v}=A^T\mathbf{u}\cdot\mathbf{v}$$

EXAMPLE 9 | Verifying that $A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T\mathbf{v}$

Suppose that

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & 1 \\ -1 & 0 & 1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix}$$

Then

$$A\mathbf{u} = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \\ 5 \end{bmatrix}$$

$$A^T\mathbf{v} = \begin{bmatrix} 1 & 2 & -1 \\ -2 & 4 & 0 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} -7 \\ 4 \\ -1 \end{bmatrix}$$

from which we obtain

$$A\mathbf{u} \cdot \mathbf{v} = 7(-2) + 10(0) + 5(5) = 11$$
$$\mathbf{u} \cdot A^T\mathbf{v} = (-1)(-7) + 2(4) + 4(-1) = 11$$

Thus, $A\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot A^T\mathbf{v}$ as guaranteed by Formula (26). We leave it for you to verify that Formula (27) also holds.